5(a)	sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop.	**B1**
5(b)(i)	$R = V / I$	**C1**
	$= (9.0 - 4.0) / 1.1 \times 10^{-2}$ $= 450\,\Omega$	**A1**
5(b)(ii)	resistance (of thermistor) = 25 (Ω) (from graph)	**C1**
	$V = E \times R_Y / (R_Y + R_X)$ $= 9 \times 25 / (25 + 450)$ **or** $I = E / R_{Total} = 9 / (25 + 450) = 1.89 \times 10^{-2}$ A $V = IR = 1.89 \times 10^{-2} \times 25$ **or** $V_{(X)} = 9 \times 450 / (25 + 450) = 8.53$ V $V = 9 - 8.53$	**C1**
	$V = 0.47$ V	**A1**
5(b)(iii)	(resistance of X increases so) the <u>total</u> resistance increases	**B1**
	current decreases	**B1**
	potential difference (across thermistor / Y) decreases	**B1**